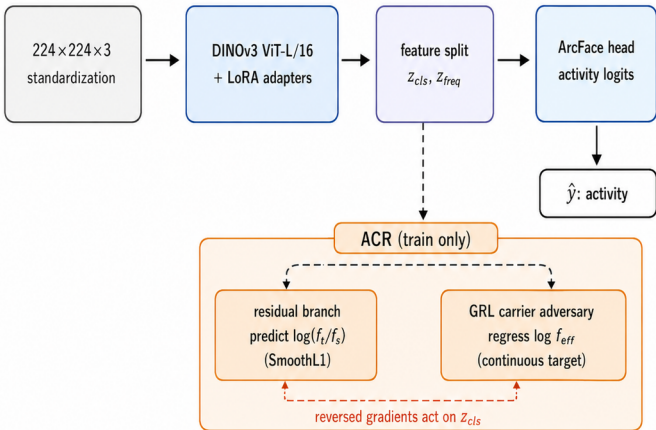
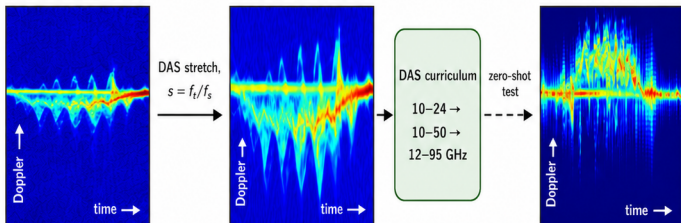


Source spectrogram
(train: 10/24 GHz)

DAS-transformed view
(24 → 77 GHz)

Held-out 77 GHz target



Final-EMA evaluation: train on 10/24 GHz only; test on held-out 77 GHz.

Macro-F1: proposed 0.832 ± 0.034 vs baseline 0.302 ± 0.031

—→ Solid: inference

- - - - -→ Dashed: train-only